

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

I Bhuvaneshwari J(192572141) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Reverse Engineering

1. System Decomposition & Analysis

Analyze an existing AI-based system and **reverse engineer its architecture**, identifying key components, data flow, and underlying algorithms.

2. Architecture Reconstruction

Given a black-box software system, **reconstruct its system architecture** and explain the interaction between modules using appropriate diagrams.

3. Algorithm Identification

Reverse engineer a machine learning application to **identify the algorithms, data processing steps, and decision-making logic** used in the system.

4. Data Flow & Process Mapping

Examine an existing distributed AI application and **map its data flow, processing pipeline, and communication between components**.

5. Improvement & Redesign

Perform reverse engineering on an intelligent system and **propose enhancements or redesign strategies** to improve scalability, efficiency, and performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. System Decomposition & Analysis

The existing AI-based system can be decomposed into three major modules:

Module 1 – Data Acquisition & Preprocessing

- Collects the NSL - KDD network intrusion dataset.
- Removes missing values and duplicate records.
- Normalizes the network traffic features.
- Splits the dataset into training and testing data.
- Produces clean data for the AI detection engine.

Module 2 – AI-Based Intrusion Detection Engine

- Receives the preprocessed network data.
- Uses Random Forest, Decision Tree, and XGBoost algorithms.
- Classifies traffic as Normal or Malicious.
- Generates intrusion detection results and security alerts.

Module 3 – Real-Time Cybersecurity Monitoring Dashboard

- Receives results from the AI detection engine.
- Displays normal and attack traffic.
- Shows graphs, statistics, attack types, and alerts.
- Generates security reports for administrators.

Overall data flow:

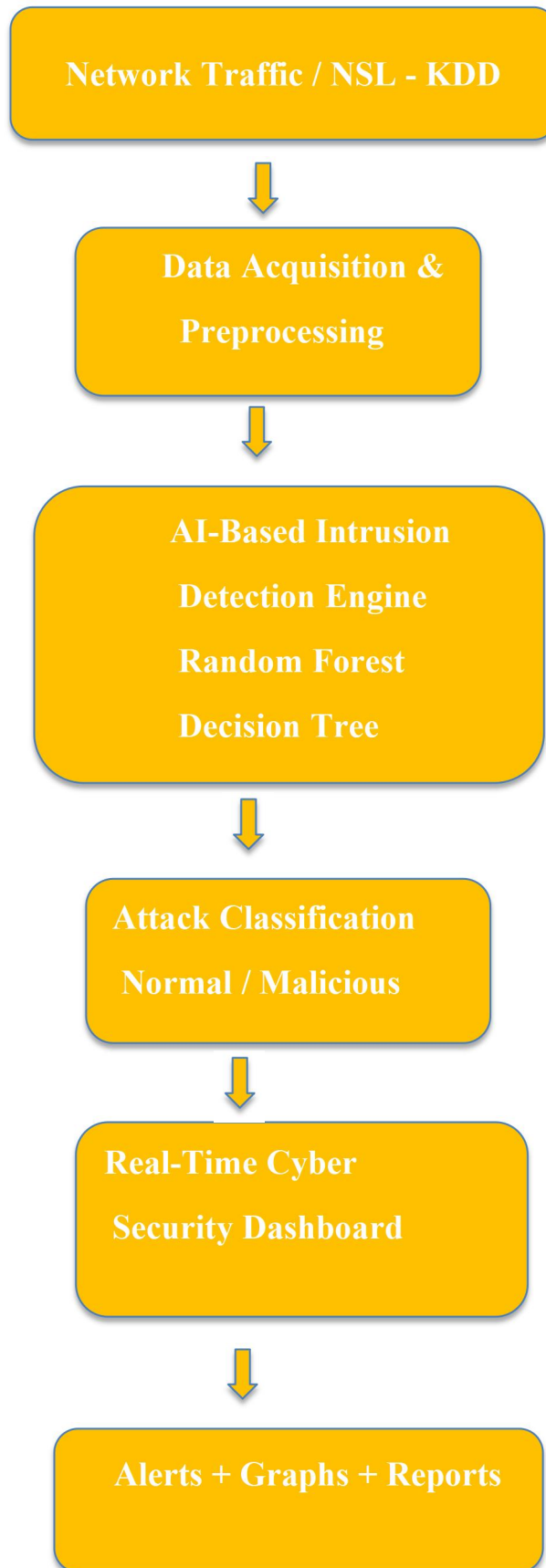
NSL - KDD Dataset → Data Preprocessing → ML Detection Engine → Attack Classification → Real-Time Dashboard → Alerts/Reports

The system is designed to continuously analyze network traffic and identify suspicious activities in real time.

2. Architecture Reconstruction

The architecture can be reconstructed from the functionality presented in the PPT.

Architecture:



Module interaction:

1. The dataset/network traffic enters the system.
2. Module 1 cleans and prepares the data.
3. The processed data is transferred to Module 2.
4. The ML models analyze the traffic.
5. The system identifies whether the traffic is normal or malicious.
6. The detection results are sent to Module 3.
7. The dashboard displays attacks, threat information, graphs, and alerts.

This reconstruction matches the PPT's three-module architecture.

3. Algorithm Identification

From the PPT, the main algorithms used by the AI-based intrusion detection engine are:

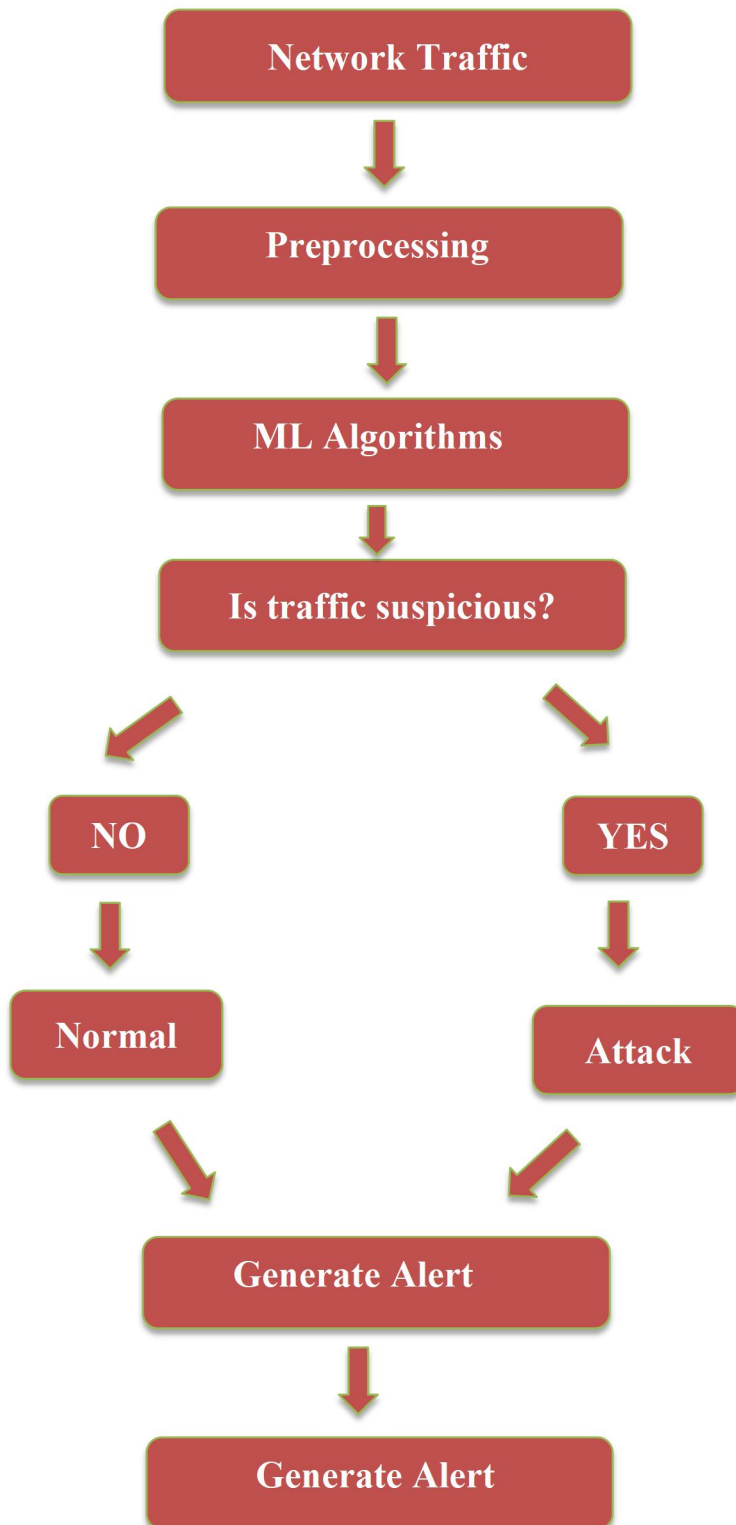
Algorithm	Role in the System
Random Forest	Classifies network traffic and helps identify malicious patterns
Decision Tree	Makes classification decisions based on network traffic features

The processing pipeline is:

Raw Network Data → Cleaning → Normalization → Train/Test Split → ML Classification → Normal/Malicious Result → Alert

The system uses the NSL - KDD dataset, which contains both benign and attack traffic, including DoS, DDoS, Brute Force, Port Scan, Botnet, Web Attacks, and Heartbleed.

Decision-making logic:



The PPT specifically states that the AI engine classifies traffic as Normal or Attack and generates security alerts.

4. Data Flow & Process Mapping

The complete data flow of the distributed AI application can be mapped as follows:

Step 1 – Data Collection

The system collects network traffic data using the NSL - KDD dataset.

Step 2 – Data Preprocessing

Missing values and duplicates are removed. Features are normalized and the data is divided into training and testing sets.

Step 3 – AI Processing

The preprocessed data is passed to the AI-based intrusion detection engine.

Step 4 – Classification

Random Forest, Decision Tree, analyze the traffic and classify it as normal or malicious.

Step 5 – Alert Generation

When malicious traffic is detected, the system generates a security alert.

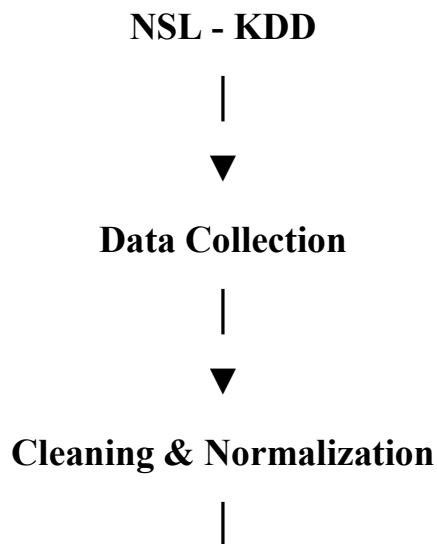
Step 6 – Dashboard Visualization

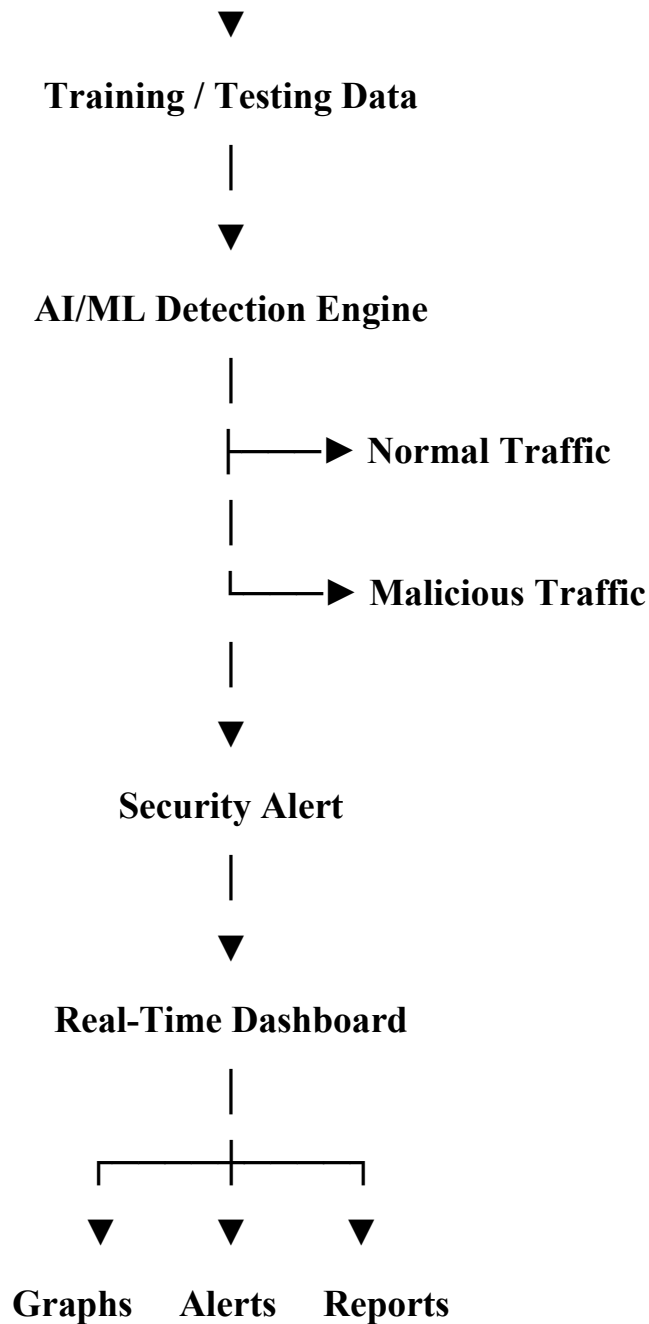
Detection results are transferred to the real-time cybersecurity dashboard.

Step 7 – Monitoring and Reporting

The administrator can view attack types, traffic statistics, alerts, graphs, and security reports.

Data Flow Diagram





5. Improvement & Redesign

Based on the future scope given in the PPT, the system can be improved in several ways:

1. Deep Learning Integration

CNN, LSTM, and Transformer models can be added to improve intrusion detection accuracy.

2. Cloud Deployment

Deploying the system on cloud infrastructure can provide scalable monitoring for larger volumes of network traffic.

3. IoT Security

The system can be extended to monitor and protect IoT and smart devices from network attacks.

4. ExplainableAI

XAI can be incorporated so that security administrators can understand why the model classified traffic as malicious.

5. Automatic Threat Response

The system can be enhanced to automatically respond to detected attacks, such as blocking malicious traffic.

6. Big Data Analytics

Big-data technologies can be incorporated to process very large volumes of network traffic in real time.